AI Engineers Learning Path · Layer 2 (Prerequisite: AI Fluent Badge)

∅ = Covered in AI Fluency Layer 1 (removed to avoid duplication) · This path builds on the foundation already established.

Section	Topic / Focus	Key Skills	Coverage	Learning Resource	Link	Dur	Cost (INR)	ML Engineer	Data Scientist	Backend Dev	DevOps
AI Foundations (Layer 1 – Reference Only)											
	∅ GenAI basics, Prompt Engineering, MS Copilot [Covered in AI Fluency Layer 1]	*Pre-requisite : Prompt engineer - Developer course*									
LLM Application Development											
	Building LLM-Powered Applications (LangChain)	LLM app frameworks, chaining, prompt templates, output parsing	Build production LLM applications: models, prompts, memory, chains	LangChain for LLM Application Development (DeepLearning.AI)	Click here	1.5	Free	Must Take	Must Take	Must Take	Advanced
	Function Calling, Tools & Structured Output	Function calling, tool selection, LCEL, structured outputs	Generate structured outputs and route tool calls using LangChain/LCEL	Functions, Tools and Agents with LangChain (DeepLearning.AI)	Click here	1.5	Free	Must Take	Optional	Must Take	Advanced
	Fine-Tuning & Model Customization	LoRA / PEFT, dataset prep, instruction tuning, evaluation	Understand when and how to fine-tune an LLM on your own data	Finetuning Large Language Models (DeepLearning.AI)	Click here	1	Free	Must Take	Must Take	Optional	Skip
RAG & Agentic Systems											
	Retrieval-Augmented Generation (RAG) Pipelines	Embeddings, chunking, vector DB integration, retrieval tuning	Design production-ready RAG systems: architecture, retrieval, evaluation	Retrieval Augmented Generation (DeepLearning.AI)	Click here	4	Free	Must Take	Must Take	Advanced	Optional
	Advanced Retrieval & RAG Evaluation	Sentence-window & auto-merging retrieval, RAG triad evaluation	Improve retrieval quality and evaluate groundedness / relevance	Building and Evaluating Advanced RAG Applications (DeepLearning.AI)	Click here	1.5	Free	Must Take	Advanced	Optional	Skip
	Agent Foundations (Hugging Face Agents Course)	Agent frameworks, tools, behaviors	How agents work; building agents with latest libraries	Hugging Face Agents Course	Click here	75	Free	Must Take	Optional	Advanced	Optional
	Multi-Agent Orchestration	Multi-agent design, coordination patterns	Design and orchestrate multi-agent AI systems	IBM: Multi-Agent Systems	Click here	1.5	Free	Optional	Optional	Advanced	Skip
	Vector Databases (Embeddings, Indexing, Similarity Search)	Embeddings, indexing, similarity search	Selecting and using vector DBs for semantic retrieval	DeepLearning.AI: Vector Databases, Embeddings, Applications	Click here	1	Free	Must Take	Must Take	Optional	Skip
	RAG + Prompt Engineering for Software Engineers (Udemy)	Applied prompt design + RAG integration patterns for dev workflows	Hands-on: combine retrieval-augmented generation with prompt engineering in a working app	AI Prompt Engineering and RAG for Software Engineers (Udemy)	Click here	2	₹ 600	Optional	Optional	Must Take	Skip
MLOps & Model Deployment											
	LLMOps: Fine-Tuning & Deployment Pipelines	Data/model versioning, tuning pipelines, safety monitoring	Design and automate an LLMOps pipeline to tune and deploy an LLM	LLMOps (DeepLearning.AI)	Click here	4	Free	Must Take	Must Take	Advanced	Optional
	Model Deployment & Serving (Docker, Kubernetes, APIs)	Containerization, endpoint serving, autoscaling	Package and serve models / LLM endpoints for production traffic	Internal Session: Model Serving & Deployment Patterns	TBD	4	Internal	Must Take	Advanced	Optional	Optional
	LLM Observability, Cost & Latency Monitoring	Tracing, evaluation dashboards, cost / perf monitoring	Monitor accuracy, latency, and cost of production LLM systems	OpenAI Model Optimization Guide	Click here	0.5	Free	Must Take	Optional	Optional	Skip
	Agent Memory Architecture (LangGraph Long-Term Memory)	Short / long-term memory, persistence patterns	Long-term agentic memory with LangGraph patterns	DeepLearning.AI: Long-Term Agentic Memory with LangGraph	Click here	1	Free	Optional	Optional	Advanced	Skip
Evaluation & Responsible AI											
	Context Engineering & Accuracy Optimization	Prompt structuring, context windows, grounding	Improve accuracy via context design, truncation, and grounding strategies	OpenAI Accuracy Optimization Guide	Click here	0.5	Free	Must Take	Optional	Optional	Skip
	RAG Evaluation with Ragas – Quickstart	Metric-based RAG evaluation: faithfulness, context precision/recall	Evaluate a RAG pipeline end-to-end using the Ragas framework	Ragas Docs – Evaluate a RAG Pipeline	Click here	1	Free	Must Take	Must Take	Optional	Skip

	RAG Evaluation with Ragas – Production Metrics	Setting up metrics to evaluate & improve RAG in production	Design a metric suite to continuously evaluate and improve a live RAG system	Ragas Docs – Evaluate & Improve RAG	Click here	1	Free	Must Take	Must Take	Optional	Skip
	Ragas for Text-to-SQL Evaluation	Experiment-driven evaluation for text-to-SQL / structured-output agents	Run structured experiments to evaluate text-to-SQL agent accuracy with Ragas	Ragas Docs – Text2SQL Evaluation	Click here	1	Free	Optional	Must Take	Optional	Skip
	Ragas – Aligning LLM-as-Judge	Calibrating an LLM-as-judge evaluator against human preference	Align automated LLM-judge scoring with human review for reliable evaluation	Ragas Docs – Align LLM-as-Judge	Click here	1	Free	Must Take	Must Take	Optional	Skip
Incedo Platforms											
	GenAI HUB	Platform awareness	AI learning platforms with learning resources and hands-on environment	GenAI HUB(session recording)	Link	1	Internal	Must Take	Must Take	Must Take	Must Take
	Lighthouse	Platform awareness	Lighthouse capabilities and business applications	Lighthouse (Internal Session)	Link	1	Internal	Optional	Optional	Must Take	Optional
	Incedo Pay	Platform awareness	Incedo Pay platform and relevant use cases	Incedo Pay (Internal Session)	Link	1	Internal	Optional	Optional	Must Take	Optional
	BrainSpark	AI platform	BrainSpark AI capabilities and project applications	BrainSpark (Internal Session)	Link	1	Internal	Optional	Optional	Must Take	Optional
	IntelliQA	AI-enabled testing	IntelliQA platform and AI-enabled testing capabilities	IntelliQA (Internal Session)	Link	1	Internal	Optional	Optional	Optional	Must Take
	DataXel	Data-driven AI	DataXel platform and data-driven AI capabilities	DataXel (Internal Session)	Link	1	Internal	Optional	Optional	Must Take	Optional
Cloud & Platform Tools											
	AWS Skill Builder (Bedrock, SageMaker)	AWS GenAI services, Bedrock, SageMaker	Skill paths on AWS GenAI; content and time vary by chosen track	AWS Skill Builder	Click here	Variable	Subscription	Must Take (AWS)	Optional	Optional	Must Take (AWS)
	Salesforce	CRM management, data analysis, automation	Sales Cloud, Service Cloud, Marketing Cloud, and beyond.	Internal Session: Model Serving & Deployment Patterns	Click Here	Variable	Internal	Must Take	Optional	Optional	Optional
	AI Automation – GitHub Prompt Action	AI workflow automation	GitHub Actions to trigger AI prompts	GitHub Marketplace: Prompt Action	Click here	0.5	Free	Optional	Optional	Optional	Optional
Hands-on & Certification											
	Capstone Project	End-to-end client-ready POC build and live demo	Own entire product life cycle - Requirement to Prod building + Client Consulting	Hands-on Exercise (SME pair build + live demo)	Internal	Self Paced	Free	Must Take	Must Take	Must Take	Must Take
	Creating Custom Agents Workshops	Workshop Series	Creation of custom agents as per requirements	Sessions recording	Click Here	3	Free	Must Take	Must Take	Must Take	Must Take
	AI SDLC	Workshop Series	Full AI SDLC from knowledge to implementation	Sessions recording	Click Here	3	Free	Must Take	Must Take	Must Take	Must Take
For relevant certifications - refer to Certification Framework											